So why is this a bad thing? The idea is that a malicious actor tricks your unsuspecting user into creating a known (to the attacker) session ID. A server that is happy to make use of a user-supplied session ID is sometimes said to be "adopting" a session ID (and this form of fixation attack is known as *session adoption*). Admittedly this is more difficult to do with cookies, which is why `session.use_only_cookies = 1` is a good idea and is the default for PHP 8.4. It is far easier to achieve with URL-based session IDs and a little social engineering. PHP 8.4 deprecates the passing of session IDs in URLs (by defaulting the ini directive to `0` and making it a non-configurable value), and PHP 9 will remove the mechanism entirely. The `session.use_only_cookies = 0` directive, together with `session.use_strict_mode = 0`, would allow the client to control the session ID on older versions, and we can demonstrate with the following code (make sure to use a version of PHP older than 8.4 for this one):

```
session_id($_GET['PHPSESSID']);
session_start();
```

```
$ curl -v http://localhost:8080/index.php?PHPSESSID=abc123
...
< Set-Cookie: PHPSESSID=abc123; path=/
```

Now imagine that your user was tricked into clicking a malicious link … the attacker now knows the session ID and can interact with it. To prevent fixation attacks, do not allow session IDs that are not created inside your application.

Disable Transparent SID Support

In the olden days of a decade or so ago, it was common to pass session IDs in URLs, and PHP would intervene with the processing of HTML elements that supported URLs (e.g., `<a href="...` `<frame src="...`). Modern PHP still contains the remnants of support for these mechanisms, known as "transparent SID support" which automagically adds session ID parameters into URLs. Obviously, a terrible idea these days and the good news is it is being expunged from PHP 8.4 onwards. Be wary of `session.use_trans_sid`, `session.trans_sid_tags` and `session.trans_sid_hosts` ini directives on older installations.

PHP 8.4 entirely deprecates the changing of `session.use_only_cookies`, `session.use_trans_sid`, `session.trans_sid_tags`, `session.trans_sid_hosts`, and `session.referer_check` ini directives, and defaults to the most secure settings.